Binary, units of data and denary–binary conversion

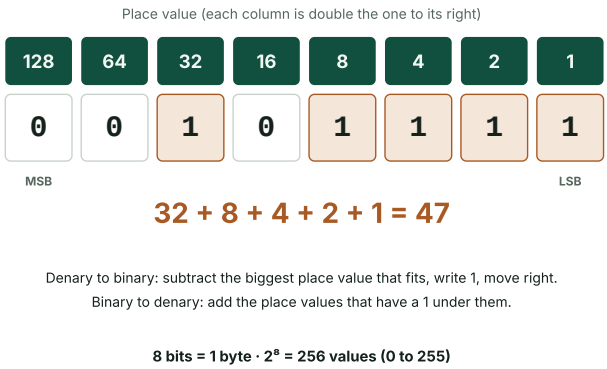
D03 FACT FILE

Specification: students should be able to

- Describe how a number is converted to a binary pattern for storage in a computer
- Demonstrate understanding of the following units of data: bit, nibble, byte, kilobyte, megabyte, gigabyte and terabyte
- Demonstrate understanding of and use number representation and convert between denary, binary and hexadecimal (denary–binary)

CCEA fact file: Digital Data

Key words: binary, denary, bit, nibble, byte, place value, kilobyte, megabyte, gigabyte, terabyte, most significant bit, least significant bit



Place values of an 8-bit byte showing 47 as 00101111

Binary patterns

A computer stores every number as a pattern of bits. Each column in binary is worth twice the column to its right: 128, 64, 32, 16, 8, 4, 2, 1 for one byte. To convert denary to binary either subtract the largest place value that fits and move right, or divide by 2 repeatedly and read the remainders upwards. To convert binary to denary, add the place values of every column that holds a 1. Always show working and write a full 8 bits.

Units

A bit is a single 0 or 1; a nibble is 4 bits; a byte is 8 bits and normally stores one character. Each larger unit is 1024 of the one below: kilobyte, megabyte, gigabyte, terabyte. With n bits there are 2^n possible patterns, so one byte can hold 256 values from 0 to 255.

T Key terms from the mark schemes

Denary
Base-10 number system.
Credited: U4 2025 Q4

Binary
Base-2 number system using 0 and 1.
Credited: U4 2022 Q6, 2025 Q4

Exam focus

Every Unit 4 paper has a conversion question: 2022 Q6(a) (54), 2023 Q7(b) (47), 2024 Q4(b)–(c) (80 and 10101010), 2025 Q4(a). Unit-matching questions on bit, nibble and byte appeared in 2024 Q4(a) and 2025 Q4(c). Marks are given for working.

Unit 4 2025 Q4	[15]	Binary conversion, addition and overflow, units, Unicode vs ASCII
Unit 4 2024 Q4	[19]	Bit, nibble, byte; denary–binary–hex conversions; binary addition and overflow; truth tables
Unit 4 2023 Q7	[13]	Bit, 47 to binary, 66 to hex, advantage of hex, binary addition and overflow
Unit 4 2022 Q6	[14]	Binary digit, 54 to binary, nibble, hex, overflow, storage capacity
Unit 4 2019 Q8	[18]	Binary, hexadecimal and its uses, binary addition and overflow, logic gates

? Check yourself

1. Explain what is meant by a bit. [2]
2. Convert the denary number 54 into an 8-bit binary number. Show your working. [2]
3. Convert the binary number 10101010 into denary. Show your working. [2]

Answers: 1. A bit is a binary digit (1) / the smallest unit of data, with a value of either 0 or 1 (off/on) (1) 2. Working, e.g. 32 + 16 + 4 + 2 = 54 under place values 128 64 32 16 8 4 2 1 (or repeated division by 2 with remainders) (1) / 00110110 (1) 3. Working: 128 + 32 + 8 + 2 (1) / = 170 (1)